

**LAPORAN PRAKTIKUM  
PEMROGRAMAN MOBILE  
MODUL 3**



**BUILD A SCROLLABLE LIST**

**Oleh:**

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**LAPORAN PRAKTIKUM PEMROGRAMAN MOBILE**  
**MODUL 3**

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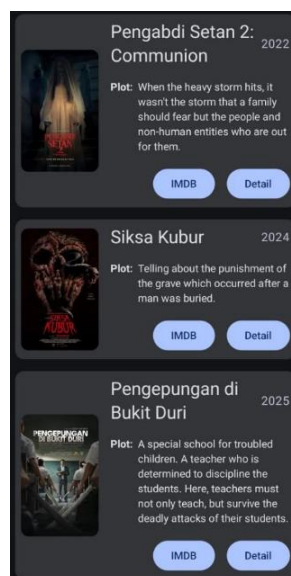
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## Soal Praktikum:

### SOAL 1

Buatlah sebuah aplikasi Android menggunakan XML dan Jetpack Compose yang dapat menampilkan list dengan ketentuan berikut:

1. List menggunakan fungsi RecyclerView (XML) dan LazyColumn (Compose)
2. List paling sedikit menampilkan 5 item. Tema item yang ingin ditampilkan bebas
3. Item pada list menampilkan teks dan gambar sesuai dengan contoh di bawah
4. Terdapat 2 button dalam list, dengan fungsi berikut:
  - a) Button pertama menggunakan intent eksplisit untuk membuka aplikasi atau browser lain
  - b) Button kedua menggunakan Navigation component untuk membuka laman detail item
5. Sudut item pada list dan gambar di dalam list melengkung atau rounded corner menggunakan Radius
6. Saat orientasi perangkat berubah/dirotasi, baik ke portrait maupun landscape, aplikasi responsif dan dapat menunjukkan list dengan baik. Data di dalam list tidak boleh hilang
7. Aplikasi menggunakan arsitektur single activity (satu activity memiliki beberapa fragment)
8. Aplikasi berbasis XML harus menggunakan ViewBinding



Gambar 1 Contoh UI List



Gambar 2 Contoh UI Detail

## A. Source Code

### Compose

#### GameDetailScreen.kt

```

1 package com.example.affirmations.ui.screens
2
3 import android.content.Intent
4 import android.net.Uri
5 import androidx.compose.foundation.background
6 import androidx.compose.foundation.layout.*
7 import androidx.compose.foundation.rememberScrollState
8 import androidx.compose.foundation.shape.RoundedCornerShape
9 import androidx.compose.foundation.verticalScroll
10 import androidx.compose.material.icons.Icons
11 import androidx.compose.material.icons.filled.ArrowBack
12 import androidx.compose.material3.*
13 import androidx.compose.runtime.Composable
14 import androidx.compose.ui.Alignment
15 import androidx.compose.ui.Modifier
16 import androidx.compose.ui.draw.clip
17 import androidx.compose.ui.layout.ContentScale
18 import androidx.compose.ui.platform.LocalContext
19 import androidx.compose.ui.text.style.TextAlign
20 import androidx.compose.ui.unit.dp
21 import androidx.navigation.NavController
22 import coil.compose.AsyncImage
23 import com.example.affirmations.data.RobloxGame

```

```

24 import com.example.affirmations.data.RobloxGamesData
25 import androidx.compose.ui.graphics.Brush
26 import androidx.compose.ui.graphics.Color
27
28 @OptIn(ExperimentalMaterial3Api::class)
29 @Composable
30 fun GameDetailScreen(
31     gameId: Int,
32     navController: NavController,
33     modifier: Modifier = Modifier
34 ) {
35     val game = RobloxGamesData.games.find { it.id == gameId }
36     val context = LocalContext.current
37
38     game?.let {
39         Scaffold(
40             topBar = {
41                 TopAppBar(
42                     title = { Text(text = game.title) },
43                     navigationIcon = {
44                         IconButton(onClick = {
45                             navController.navigateUp() }) {
46                             Icon(Icons.Default.ArrowBack,
47                                 contentDescription = "Back")
48                         }
49                     },
50                     colors = TopAppBarDefaults.topAppBarColors(
51                         containerColor = MaterialTheme.colorScheme.primaryContainer,
52                         titleContentColor = MaterialTheme.colorScheme.onPrimaryContainer
53                     )
54                 )
55             }
56         ) { paddingValues ->
57             Column(
58                 modifier = modifier
59                 .fillMaxSize()
60                 .padding(paddingValues)
61                 .verticalScroll(rememberScrollState())
62             ) {
63                 Box(
64                     modifier = Modifier
65                     .fillMaxWidth()
66                     .height(250.dp)
67                 ) {

```



```

67         AsyncImage(
68             model = game.imageUrl,
69             contentDescription = game.title,
70             modifier = Modifier.fillMaxSize(),
71             contentScale = ContentScale.Crop
72         )
73
74         Box(
75             modifier = Modifier
76                 .fillMaxSize()
77                 .background(
78                     brush = Brush.verticalGradient(
79                         colors = listOf(
80                             Color.Transparent,
81                             Color.Black.copy(alpha =
82 0.7f)
83                         )
84                     )
85                 )
86         )
87
88         Column(
89             modifier = Modifier
90                 .align(Alignment.BottomStart)
91                 .padding(16.dp)
92         ) {
93             Text(
94                 text = game.title,
95                 style =
96                 MaterialTheme.typography.headlineMedium,
97                 color = Color.White
98             )
99             Text(
100                 text = game.playersCount,
101                 style =
102                 MaterialTheme.typography.titleMedium,
103                 color = Color.White.copy(alpha = 0.8f)
104             )
105         }
106
107         Column(
108             modifier = Modifier
109                 .padding(16.dp)
110                 .fillMaxWidth()
111         ) {

```

```

110
111         ElevatedCard(
112             modifier = Modifier.fillMaxWidth(),
113             colors = CardDefaults.elevatedCardColors(
114                 containerColor
115                 MaterialTheme.colorScheme.surfaceVariant
116             )
117         ) {
118             Row(
119                 modifier = Modifier
120                     .fillMaxWidth()
121                     .padding(16.dp),
122                 horizontalArrangement
123                 Arrangement.SpaceEvenly
124             ) {
125                 InfoColumn("Genre", game.genre)
126                 VerticalDivider()
127                 InfoColumn("Rating", game.rating)
128                 VerticalDivider()
129                 InfoColumn("Released",
130                     game.releaseDate)
131             }
132
133             Spacer(modifier = Modifier.height(16.dp))
134
135             Text(
136                 text = "About",
137                 style
138                 MaterialTheme.typography.titleLarge,
139                 color
140                 MaterialTheme.colorScheme.onSurface
141             )
142             Spacer(modifier = Modifier.height(8.dp))
143             Text(
144                 text = game.description,
145                 style
146                 MaterialTheme.typography.bodyLarge,
147                 color
148                 MaterialTheme.colorScheme.onSurfaceVariant
149             )
150
151             Spacer(modifier = Modifier.height(16.dp))
152
153             Text(

```

|     |                                                     |   |
|-----|-----------------------------------------------------|---|
| 149 | text = "Developer",                                 |   |
| 150 | style                                               | = |
| 151 | MaterialTheme.typography.titleLarge,                |   |
| 152 | color                                               | = |
| 153 | MaterialTheme.colorScheme.onSurface                 |   |
| 154 | )                                                   |   |
| 155 | Spacer(modifier = Modifier.height(8.dp))            |   |
| 156 | Text(                                               |   |
|     | text = game.developer,                              |   |
| 157 | style                                               | = |
|     | MaterialTheme.typography.bodyLarge,                 |   |
| 158 | color                                               | = |
| 159 | MaterialTheme.colorScheme.onSurfaceVariant          |   |
| 160 | )                                                   |   |
| 161 |                                                     |   |
| 162 | Spacer(modifier = Modifier.height(24.dp))           |   |
| 163 |                                                     |   |
|     | Button(                                             |   |
| 164 | onClick = {                                         |   |
| 165 | val intent                                          | = |
| 166 | Intent(Intent.ACTION_VIEW, Uri.parse(game.gameUrl)) |   |
| 167 | context.startActivity(intent)                       |   |
| 168 | },                                                  |   |
| 169 | modifier = Modifier.fillMaxWidth(),                 |   |
| 170 | colors = ButtonDefaults.buttonColors(               |   |
| 171 | containerColor                                      | = |
|     | MaterialTheme.colorScheme.primary                   |   |
| 172 | )                                                   |   |
|     | ) {                                                 |   |
| 173 | Text(                                               |   |
| 174 | text = "Play Now",                                  |   |
| 175 | style                                               | = |
| 176 | MaterialTheme.typography.titleMedium                |   |
| 177 | )                                                   |   |
|     | }                                                   |   |
| 178 | }                                                   |   |
| 179 | }                                                   |   |
| 180 | }                                                   |   |
| 181 | } ?: run {                                          |   |
| 182 | Box(                                                |   |
|     | modifier = Modifier                                 |   |
| 183 | .fillMaxSize()                                      |   |
| 184 | .padding(16.dp)                                     |   |
| 185 | ) {                                                 |   |
| 186 | Text("Game not found")                              |   |
|     | }                                                   |   |

|     |                                                                   |
|-----|-------------------------------------------------------------------|
| 187 | }                                                                 |
| 188 | }                                                                 |
| 189 |                                                                   |
| 190 | @Composable                                                       |
| 191 | private fun InfoColumn(                                           |
| 192 | label: String,                                                    |
| 193 | value: String,                                                    |
| 194 | modifier: Modifier = Modifier                                     |
| 195 | ) {                                                               |
| 196 | Column(                                                           |
| 197 | horizontalAlignment = Alignment.CenterHorizontally,               |
| 198 | modifier = modifier                                               |
| 199 | ) {                                                               |
| 200 | Text(                                                             |
| 201 | text = label,                                                     |
| 202 | style = MaterialTheme.typography.labelMedium,                     |
| 203 | color = MaterialTheme.colorScheme.onSurfaceVariant                |
| 204 | )                                                                 |
| 205 | Spacer(modifier = Modifier.height(4.dp))                          |
| 206 | Text(                                                             |
| 207 | text = value,                                                     |
| 208 | style = MaterialTheme.typography.bodyMedium,                      |
| 209 | color = MaterialTheme.colorScheme.onSurface,                      |
| 210 | textAlign = TextAlign.Center                                      |
| 211 | )                                                                 |
| 212 | }                                                                 |
| 213 | }                                                                 |
| 214 |                                                                   |
| 215 | @Composable                                                       |
| 216 | private fun VerticalDivider(                                      |
| 217 | modifier: Modifier = Modifier                                     |
| 218 | ) {                                                               |
| 219 | Box(                                                              |
| 220 | modifier = modifier                                               |
| 221 | .width(1.dp)                                                      |
| 222 | .height(32.dp)                                                    |
| 223 |                                                                   |
| 224 | .background(MaterialTheme.colorScheme.onSurfaceVariant.copy(alpha |
| 225 | = 0.2f))                                                          |
| 226 | )                                                                 |
| 227 | }                                                                 |
| 228 |                                                                   |

Tabel 1 Source Code Jawaban Soal 1

## GameListScreen.kt

|   |                                             |
|---|---------------------------------------------|
| 1 | package com.example.affirmations.ui.screens |
| 2 |                                             |

```

3 import android.content.Intent
4 import android.net.Uri
5 import androidx.compose.foundation.layout.*
6 import androidx.compose.foundation.lazy.LazyColumn
7 import androidx.compose.foundation.lazy.items
8 import androidx.compose.foundation.shape.RoundedCornerShape
9 import androidx.compose.material3.*
10 import androidx.compose.runtime.Composable
11 import androidx.compose.ui.Alignment
12 import androidx.compose.ui.Modifier
13 import androidx.compose.ui.draw.clip
14 import androidx.compose.ui.layout.ContentScale
15 import androidx.compose.ui.platform.LocalContext
16 import androidx.compose.ui.text.style.TextOverflow
17 import androidx.compose.ui.unit.dp
18 import androidx.navigation.NavController
19 import coil.compose.AsyncImage
20 import com.example.affirmations.data.RobloxGamesData
21 import com.example.affirmations.navigation.Screen
22
23 @Composable
24 fun GamesListScreen(
25     navController: NavController,
26     modifier: Modifier = Modifier
27 ) {
28     LazyColumn(
29         modifier = modifier.fillMaxSize(),
30         contentPadding = PaddingValues(16.dp),
31         verticalArrangement = Arrangement.spacedBy(16.dp)
32     ) {
33         items(RobloxGamesData.games) { game ->
34             val context = LocalContext.current
35             GameCard(
36                 game = game,
37                 onPlayClick = {
38                     val intent = Intent(Intent.ACTION_VIEW,
39                         Uri.parse(game.gameUrl))
40                     context.startActivity(intent)
41                 },
42                 onDetailsClick = {
43                     navController.navigate(Screen.GameDetail.createRoute(game.id))
44                 }
45             )
46         }
47     }
48 }

```

```

48 @OptIn(ExperimentalMaterial3Api::class)
49 @Composable
50 private fun GameCard(
51     game: com.example.affirmations.data.RobloxGame,
52     onPlayClick: () -> Unit,
53     onDetailsClick: () -> Unit,
54     modifier: Modifier = Modifier
55 ) {
56     Card(
57         modifier = modifier.fillMaxWidth(),
58         shape = RoundedCornerShape(16.dp)
59     ) {
60         Row(
61             modifier = Modifier
62                 .padding(16.dp)
63                 .fillMaxWidth(),
64             verticalAlignment = Alignment.CenterVertically
65         ) {
66             AsyncImage(
67                 model = game.imageUrl,
68                 contentDescription = game.title,
69                 modifier = Modifier
70                     .size(width = 120.dp, height = 120.dp)
71                     .clip(RoundedCornerShape(12.dp)),
72                 contentScale = ContentScale.Crop
73             )
74
75             Spacer(modifier = Modifier.width(16.dp))
76
77             Column(
78                 modifier = Modifier.weight(1f)
79             ) {
80                 Text(
81                     text = game.title,
82                     style
83                     =
84                     MaterialTheme.typography.titleLarge
85                 )
86
87                 Spacer(modifier = Modifier.height(4.dp))
88
89                 Text(
90                     text = game.playersCount,
91                     style
92                     =
93                     MaterialTheme.typography.titleMedium,
94                     color = MaterialTheme.colorScheme.primary
95                 )

```

|     |                                            |   |
|-----|--------------------------------------------|---|
| 92  |                                            |   |
| 93  | Spacer(modifier = Modifier.height(4.dp))   |   |
| 94  |                                            |   |
| 95  | Text(                                      |   |
|     | text = "\${game.genre}",                   |   |
| 96  | style                                      | = |
| 97  | MaterialTheme.typography.bodyMedium,       |   |
| 98  | color                                      | = |
| 99  | MaterialTheme.colorScheme.onSurfaceVariant |   |
| 100 | )                                          |   |
| 101 |                                            |   |
| 102 | Text(                                      |   |
|     | text = "\${game.rating}",                  |   |
| 103 | style                                      | = |
| 104 | MaterialTheme.typography.bodyMedium,       |   |
| 105 | color                                      | = |
| 106 | MaterialTheme.colorScheme.onSurfaceVariant |   |
| 107 | )                                          |   |
| 108 |                                            |   |
| 109 | Spacer(modifier = Modifier.height(8.dp))   |   |
| 110 |                                            |   |
|     | Row(                                       |   |
| 111 | modifier = Modifier.fillMaxWidth(),        |   |
|     | horizontalArrangement                      | = |
| 112 | Arrangement.spacedBy(8.dp)                 |   |
| 113 | ) {                                        |   |
| 114 | Button(                                    |   |
| 115 | onClick = onPlayClick,                     |   |
| 116 | modifier = Modifier.weight(1f),            |   |
|     | colors = ButtonDefaults.buttonColors(      |   |
| 117 | containerColor                             | = |
|     | MaterialTheme.colorScheme.primary,         |   |
| 118 | contentColor                               | = |
| 119 | MaterialTheme.colorScheme.onPrimary        |   |
| 120 | )                                          |   |
| 121 | ) {                                        |   |
| 122 | Text(                                      |   |
| 123 | text = "Play Now",                         |   |
| 124 | maxLines = 1,                              |   |
|     | overflow = TextOverflow.Ellipsis           |   |
| 125 | )                                          |   |
| 126 | }                                          |   |
| 127 |                                            |   |
| 128 | Button(                                    |   |
| 129 | onClick = onDetailsClick,                  |   |
| 130 | modifier = Modifier.weight(1f),            |   |
|     | colors = ButtonDefaults.buttonColors(      |   |

|     |                                       |   |
|-----|---------------------------------------|---|
| 131 | containerColor                        | = |
|     | MaterialTheme.colorScheme.secondary,  |   |
| 132 | contentColor                          | = |
|     | MaterialTheme.colorScheme.onSecondary |   |
| 133 | )                                     |   |
| 134 | ) {                                   |   |
| 135 | Text(                                 |   |
| 136 | text = "View Details",                |   |
| 137 | maxLines = 1,                         |   |
| 138 | overflow = TextOverflow.Ellipsis      |   |
| 139 | )                                     |   |
| 140 | }                                     |   |
| 141 | }                                     |   |
| 142 | }                                     |   |
| 143 | }                                     |   |
| 144 | }                                     |   |
| 145 | }                                     |   |

Tabel 2 Source Code Jawaban Soal 1

### RobloxGame.kt

|    |                                                                |
|----|----------------------------------------------------------------|
| 1  | package com.example.modul4compose.data                         |
| 2  |                                                                |
| 3  | data class RobloxGame(                                         |
| 4  | val id: Int,                                                   |
| 5  | val title: String,                                             |
| 6  | val description: String,                                       |
| 7  | val imageUrl: String,                                          |
| 8  | val playersCount: String,                                      |
| 9  | val gameUrl: String,                                           |
| 10 | val genre: String,                                             |
| 11 | val rating: String,                                            |
| 12 | val releaseDate: String,                                       |
| 13 | val developer: String                                          |
| 14 | )                                                              |
| 15 |                                                                |
| 16 | object RobloxGamesData {                                       |
| 17 | val games = listOf(                                            |
| 18 | RobloxGame(                                                    |
| 19 | 1,                                                             |
| 20 | "Adopt Me!",                                                   |
| 21 | "A role-playing game where players can adopt and               |
|    | raise pets, decorate homes, and interact with other players.   |
|    | Features include pet trading, house customization, and mini-   |
|    | games.",                                                       |
| 22 | "https://tr.rbxcdn.com/180DAY-                                 |
|    | 23a22e242c44352464dbca03bbcb189a/768/432/Image/Webp/noFilter", |
| 23 | "500K+ Active Players",                                        |



```

24         "https://www.roblox.com/games/920587237",
25         "Role-Playing",
26         "4.5/5",
27         "July 2017",
28         "DreamCraft"
29     ),
30     RobloxGame(
31         2,
32         "Brookhaven RP",
33         "A roleplaying game where players can live, work,
and socialize in a virtual city. Features include house
ownership, vehicle system, and various jobs to choose from.",
34         "https://tr.rbxcdn.com/180DAY-
d743ef79fbc0a90f9aea0bdf740cd00c/768/432/Image/Webp/noFilter",
35         "450K+ Active Players",
36         "https://www.roblox.com/games/492492222",
37         "Role-Playing",
38         "4.3/5",
39         "April 2020",
40         "Wolfpaq"
41     ),
42     RobloxGame(
43         3,
44         "Blox Fruits",
45         "An action-adventure game inspired by One Piece
where players can train, fight, and collect Devil Fruits.
Features include PvP combat, fruit hunting, and character
progression.",
46         "https://tr.rbxcdn.com/180DAY-
37cc49a6bdb03b5a4394db84ff264fe5/768/432/Image/Webp/noFilter",
47         "400K+ Active Players",
48         "https://www.roblox.com/games/2753915549",
49         "Action-Adventure",
50         "4.7/5",
51         "January 2019",
52         "Gamer Robot Inc"
53     ),
54     RobloxGame(
55         4,
56         "Pet Simulator X",
57         "Collect pets, hatch eggs, and explore various
worlds while becoming the ultimate pet master. Features include
trading system, rare pets collection, and world exploration.",
58         "https://tr.rbxcdn.com/180DAY-
384dc534a8bd24e4c6217056c1d4ad4f/768/432/Image/Webp/noFilter",
59         "300K+ Active Players",
60         "https://www.roblox.com/games/6284583030",

```

|    |                                                                                                                                                                                              |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 61 | "Simulation",                                                                                                                                                                                |
| 62 | "4.4/5",                                                                                                                                                                                     |
| 63 | "July 2021",                                                                                                                                                                                 |
| 64 | "BIG Games"                                                                                                                                                                                  |
| 65 | ),                                                                                                                                                                                           |
| 66 | RobloxGame(                                                                                                                                                                                  |
| 67 | 5,                                                                                                                                                                                           |
| 68 | "Murder Mystery 2",                                                                                                                                                                          |
| 69 | "A thrilling game where players are assigned roles<br>of innocent, sheriff, or murderer in each round. Features<br>include weapon collecting, trading system, and competitive<br>gameplay.", |
| 70 | "https://tr.rbxcdn.com/180DAY-<br>c929c9ba9069190914f60bbfe47b6cb9/768/432/Image/Webp/noFilter",                                                                                             |
| 71 | "250K+ Active Players",                                                                                                                                                                      |
| 72 | "https://www.roblox.com/games/142823291",                                                                                                                                                    |
| 73 | "Action",                                                                                                                                                                                    |
| 74 | "4.6/5",                                                                                                                                                                                     |
| 75 | "January 2014",                                                                                                                                                                              |
| 76 | "Nikilis"                                                                                                                                                                                    |
| 77 | )                                                                                                                                                                                            |
| 78 | )                                                                                                                                                                                            |
| 79 | }                                                                                                                                                                                            |

Tabel 3 Source Code Jawaban Soal 1

## NavGraph.kt

|    |                                                      |
|----|------------------------------------------------------|
| 1  | package com.example.modul4compose.navigation         |
| 2  |                                                      |
| 3  | sealed class Screen(val route: String) {             |
| 4  | object GamesList : Screen("games_list")              |
| 5  | object GameDetail : Screen("game_detail/{gameId}") { |
| 6  | fun createRoute(gameId: Int) =                       |
| 7  | "game_detail/\$gameId"                               |
| 8  | }                                                    |
| 9  | }                                                    |
| 10 | object NavArguments {                                |
| 11 | const val GAME_ID = "gameId"                         |
| 12 | }                                                    |

Tabel 4 Source Code Jawaban Soal 1

## MainActivity.kt (Compose)

|   |                                  |
|---|----------------------------------|
| 1 | package com.example.affirmations |
| 2 |                                  |
| 3 | import android.os.Bundle         |

```

4 import androidx.activity.ComponentActivity
5 import androidx.activity.compose.setContent
6 import androidx.compose.foundation.layout.fillMaxSize
7 import androidx.compose.material3.MaterialTheme
8 import androidx.compose.material3.Surface
9 import androidx.compose.ui.Modifier
10 import androidx.navigation.NavType
11 import androidx.navigation.compose.NavHost
12 import androidx.navigation.compose.composable
13 import androidx.navigation.compose.rememberNavController
14 import androidx.navigation.navArgument
15 import com.example.affirmations.navigation.NavArguments
16 import com.example.affirmations.navigation.Screen
17 import
18 com.example.affirmations.ui.screens.GameDetailScreen
19 import com.example.affirmations.ui.screens.GamesListScreen
20 import com.example.affirmations.ui.theme.AffirmationsTheme
21
22 class MainActivity : ComponentActivity() {
23     override fun onCreate(savedInstanceState: Bundle?) {
24         super.onCreate(savedInstanceState)
25         setContent {
26             AffirmationsTheme {
27                 Surface(
28                     modifier = Modifier.fillMaxSize(),
29                     color
30 MaterialTheme.colorScheme.background
31                 ) {
32                     val navController
33 rememberNavController()
34
35                     NavHost(
36                         navController = navController,
37                         startDestination
38 Screen.GamesList.route
39                     ) {
40 composable(Screen.GamesList.route) {
41                     GamesListScreen(navController
42 = navController)
43                 }
44
45                 composable(
46                     route
47 Screen.GameDetail.route,
48                     arguments = listOf(

```

|    |                                                            |
|----|------------------------------------------------------------|
|    | navArgument (NavArguments.GAME_ID) {                       |
| 44 | type = NavType.IntType                                     |
| 45 | }                                                          |
| 46 | )                                                          |
| 47 | ) { backStackEntry ->                                      |
| 48 | val gameId =                                               |
|    | backStackEntry.arguments?.getInt (NavArguments.GAME_ID) ?: |
|    | 0                                                          |
| 49 | GameDetailScreen (                                         |
| 50 | gameId = gameId,                                           |
| 51 | navController =                                            |
| 52 | navController                                              |
| 53 | )                                                          |
| 54 | }                                                          |
| 55 | }                                                          |
| 56 | }                                                          |
| 57 | }                                                          |
| 58 | }                                                          |
| 59 | }                                                          |
| 60 | }                                                          |

Tabel 5 Source Code Jawaban Soal 1

## XML

### activity\_main.xml

|    |                                                             |
|----|-------------------------------------------------------------|
| 1  | <?xml version="1.0" encoding="utf-8"?>                      |
| 2  | <androidx.constraintlayout.widget.ConstraintLayout          |
| 3  | xmlns:android="http://schemas.android.com/apk/res/android"  |
| 4  | xmlns:app="http://schemas.android.com/apk/res-auto"         |
| 5  | android:layout_width="match_parent"                         |
| 6  | android:layout_height="match_parent">                       |
| 7  |                                                             |
| 8  | <androidx.fragment.app.FragmentContainerView                |
| 9  | android:id="@+id/nav_host_fragment"                         |
| 10 | android:name="androidx.navigation.fragment.NavHostFragment" |
| 11 | android:layout_width="match_parent"                         |
| 12 | android:layout_height="match_parent"                        |
| 13 | app:defaultNavHost="true"                                   |
| 14 | app:navGraph="@navigation/nav_graph" />                     |
| 15 |                                                             |
| 16 | </androidx.constraintlayout.widget.ConstraintLayout>        |

Tabel 6Source Code Jawaban Soal 1

### fragment\_roblox\_game\_detail.xml

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.core.widget.NestedScrollView
3  xmlns:android="http://schemas.android.com/apk/res/an
4  droid"
5      xmlns:app="http://schemas.android.com/apk/res-
6  auto"
7      xmlns:tools="http://schemas.android.com/tools"
8      android:layout_width="match_parent"
9      android:layout_height="match_parent"
10     android:fillViewport="true">
11
12
13     <androidx.constraintlayout.widget.ConstraintLayout
14         android:layout_width="match_parent"
15         android:layout_height="wrap_content">
16
17
18         <ImageView
19             android:id="@+id/gameHeaderImage"
20             android:layout_width="match_parent"
21             android:layout_height="200dp"
22             android:scaleType="centerCrop"
23
24             app:layout_constraintTop_toTopOf="parent"
25             tools:src="@drawable/adopt_me"/>
26
27
28         <TextView
29             android:id="@+id/gameTitle"
30             android:layout_width="match_parent"
31             android:layout_height="wrap_content"
32             android:layout_marginHorizontal="16dp"
33             android:layout_marginTop="16dp"
34             android:textAppearance="@style/TextAppearance.Materi
35             alComponents.Headline5"
36             android:textStyle="bold"
37             app:layout_constraintTop_toBottomOf="@id/gameHeaderI
38             mage"
39             tools:text="Adopt Me!"/>
40
41
42     <com.google.android.material.card.MaterialCardView
43         android:id="@+id/infoContainer"

```

```

44         android:layout_width="match_parent"
45         android:layout_height="wrap_content"
46         android:layout_margin="16dp"
47         app:cardCornerRadius="8dp"
48         app:cardElevation="4dp"
49
50     app:layout_constraintTop_toBottomOf="@id/gameTitle">
51
52         <LinearLayout
53             android:layout_width="match_parent"
54             android:layout_height="wrap_content"
55             android:orientation="horizontal"
56             android:padding="16dp"
57             android:weightSum="3">
58
59
60             <LinearLayout
61                 android:layout_width="0dp"
62
63                 android:layout_height="wrap_content"
64                 android:layout_weight="1"
65                 android:gravity="center"
66                 android:orientation="vertical">
67
68                 <TextView
69
70                 android:layout_width="wrap_content"
71
72                 android:layout_height="wrap_content"
73                     android:text="Genre"
74
75                 android:textAppearance="@style/TextAppearance.MaterialComponents.Caption"/>
76
77                 <TextView
78                     android:id="@+id/gameGenre"
79
80                 android:layout_width="wrap_content"
81
82                 android:layout_height="wrap_content"
83
84                 android:layout_marginTop="4dp"
85
86                 android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"
87
88                 tools:text="Role-Playing"/>
89             </LinearLayout>

```

```

        <LinearLayout
            android:layout_width="0dp"

android:layout_height="wrap_content"
            android:layout_weight="1"
            android:gravity="center"
            android:orientation="vertical">

            <TextView

android:layout_width="wrap_content"

android:layout_height="wrap_content"
                android:text="Rating"

android:textAppearance="@style/TextAppearance.MaterialComponents.Caption"/>

                <TextView
                    android:id="@+id/gameRating"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:layout_marginTop="4dp"

android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"
                    tools:text="4.5/5"/>
            </LinearLayout>

        <LinearLayout
            android:layout_width="0dp"

android:layout_height="wrap_content"
            android:layout_weight="1"
            android:gravity="center"
            android:orientation="vertical">

            <TextView

android:layout_width="wrap_content"

```

```

        android:layout_height="wrap_content"
            android:text="Released"

        android:textAppearance="@style/TextAppearance.MaterialComponents.Caption"/>

        <TextView

        android:id="@+id/gameReleased"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_marginTop="4dp"

        android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"
            tools:text="July 2017"/>
        </LinearLayout>

    </LinearLayout>

</com.google.android.material.card.MaterialCardView>

    <TextView
        android:id="@+id/aboutTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginTop="24dp"
        android:text="About"

        android:textAppearance="@style/TextAppearance.MaterialComponents.Headline6"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintTop_toBottomOf="@id/infoContainer"/>

    <TextView
        android:id="@+id/gameDescription"
        android:layout_width="match_parent"

```



```

        android:layout_height="wrap_content"
        android:layout_marginHorizontal="16dp"
        android:layout_marginTop="8dp"

        android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"

        app:layout_constraintTop_toBottomOf="@id/aboutTitle"
            tools:text="A role-playing game where
players can adopt
and raise pets, decorate homes, and interact with
other players.
Features include pet trading, house customization,
and mini-games."/>

        <TextView
            android:id="@+id/developerTitle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="16dp"
            android:layout_marginTop="24dp"
            android:text="Developer"

            android:textAppearance="@style/TextAppearance.MaterialComponents.Headline6"

            app:layout_constraintStart_toStartOf="parent"

            app:layout_constraintTop_toBottomOf="@id/gameDescription"
            />

        <TextView
            android:id="@+id/gameDeveloper"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginHorizontal="16dp"
            android:layout_marginTop="8dp"

            android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"

            app:layout_constraintTop_toBottomOf="@id/developerTitle"
                tools:text="DreamCraft"/>

```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> &lt;com.google.android.material.button.MaterialButton     android:id="@+id/playNowButton"     android:layout_width="match_parent"     android:layout_height="wrap_content"     android:layout_margin="16dp"     android:padding="12dp"     android:text="Play Now"     android:textSize="16sp"      app:layout_constraintBottom_toBottomOf="parent"      app:layout_constraintTop_toBottomOf="@id/gameDeveloper"      app:layout_constraintVertical_bias="1.0"/&gt;  &lt;/androidx.constraintlayout.widget.ConstraintLayout&gt;  &lt;/androidx.core.widget.NestedScrollView&gt; </pre> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Tabel 7.Source Code Jawaban Soal 1

### fragment\_roblox\_game\_list.xml

|    |                                                                      |
|----|----------------------------------------------------------------------|
| 1  | <?xml version="1.0" encoding="utf-8"?>                               |
| 2  | <androidx.constraintlayout.widget.ConstraintLayout                   |
| 3  | xmlns:android="http://schemas.android.com/apk/res/android"           |
| 4  | xmlns:app="http://schemas.android.com/apk/res-auto"                  |
| 5  | android:layout_width="match_parent"                                  |
| 6  | android:layout_height="match_parent">                                |
| 7  |                                                                      |
| 8  | <androidx.recyclerview.widget.RecyclerView                           |
| 9  | android:id="@+id/gamesRecyclerView"                                  |
| 10 | android:layout_width="match_parent"                                  |
| 11 | android:layout_height="match_parent"                                 |
| 12 | android:clipToPadding="false"                                        |
| 13 | android:padding="8dp"                                                |
| 14 | app:layoutManager="androidx.recyclerview.widget.LinearLayoutManager" |
| 15 | app:layout_constraintBottom_toBottomOf="parent"                      |
| 16 | app:layout_constraintEnd_toEndOf="parent"                            |

|    |                                                      |
|----|------------------------------------------------------|
| 17 |                                                      |
| 18 | app:layout_constraintStart_toStartOf="parent"        |
| 19 | app:layout_constraintTop_toTopOf="parent" />         |
| 20 |                                                      |
|    | </androidx.constraintlayout.widget.ConstraintLayout> |

Tabel 8 Source Code Jawaban Soal 1

### item\_roblox\_game.xml

|    |                                                            |
|----|------------------------------------------------------------|
| 1  | <?xml version="1.0" encoding="utf-8"?>                     |
| 2  | <com.google.android.material.card.MaterialCardView         |
| 3  | xmlns:android="http://schemas.android.com/apk/res/android" |
| 4  | xmlns:app="http://schemas.android.com/apk/res-auto"        |
| 5  | xmlns:tools="http://schemas.android.com/tools"             |
| 6  | android:layout_width="match_parent"                        |
| 7  | android:layout_height="wrap_content"                       |
| 8  | android:layout_margin="8dp"                                |
| 9  | app:cardCornerRadius="12dp"                                |
| 10 | app:cardElevation="4dp">                                   |
| 11 |                                                            |
| 12 |                                                            |
| 13 | <androidx.constraintlayout.widget.ConstraintLayout         |
|    | android:layout_width="match_parent"                        |
| 14 | android:layout_height="wrap_content"                       |
|    | android:padding="8dp">                                     |
| 15 |                                                            |
| 16 | <com.google.android.material.imageview.ShapeableImageView  |
| 17 | android:id="@+id/gameImage"                                |
| 18 | android:layout_width="120dp"                               |
| 19 | android:layout_height="120dp"                              |
| 20 | android:scaleType="centerCrop"                             |
|    | app:layout_constraintStart_toStartOf="parent"              |
|    | app:layout_constraintTop_toTopOf="parent"                  |
|    | app:shapeAppearanceOverlay="@style/RoundedCornerImage"     |
|    | tools:src="@drawable/placeholder"/>                        |
|    | <TextView                                                  |
|    | android:id="@+id/gameTitle"                                |
|    | android:layout_width="0dp"                                 |

```

        android:layout_height="wrap_content"
        android:layout_marginStart="12dp"
        android:textSize="18sp"
        android:textStyle="bold"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintStart_toEndOf="@id/gameImage"

app:layout_constraintTop_toTopOf="parent"
    tools:text="Adopt Me!"/>

    <TextView
        android:id="@+id/gameGenre"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="12dp"
        android:layout_marginTop="4dp"
        android:textSize="14sp"

app:layout_constraintStart_toEndOf="@id/gameImage"

app:layout_constraintTop_toBottomOf="@id/gameTitle"
    tools:text="Role-Playing"/>

    <TextView
        android:id="@+id/gamePlayers"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="12dp"
        android:textColor="#674fa3"
        android:textSize="14sp"

app:layout_constraintStart_toEndOf="@id/gameGenre"

app:layout_constraintTop_toTopOf="@id/gameGenre"
    tools:text="500K+ Players" />

    <TextView
        android:id="@+id/gameRating"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="12dp"
        android:layout_marginTop="4dp"
        android:textSize="14sp"

app:layout_constraintStart_toEndOf="@id/gameImage"

```

```

app:layout_constraintTop_toBottomOf="@id/gameGenre"
    tools:text="Rating: 4.5/5"/>

    <Button
        android:id="@+id/playNowButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:backgroundTint="#674fa3"
        android:text="Play Now"
        android:textSize="12sp"

app:layout_constraintStart_toStartOf="@id/gameGenre"

app:layout_constraintTop_toBottomOf="@id/gameRating"
/>

    <Button
        android:id="@+id/viewDetailsButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:backgroundTint="#674fa3"
        android:text="View Details"
        android:textSize="12sp"

app:layout_constraintStart_toEndOf="@id/playNowButton"
n"

app:layout_constraintTop_toTopOf="@id/playNowButton"
/>

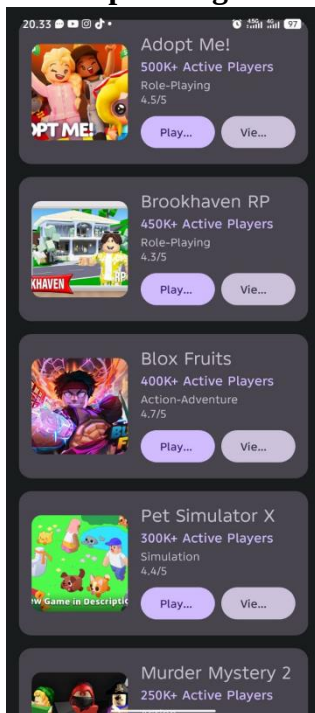
</androidx.constraintlayout.widget.ConstraintLayout>

</com.google.android.material.card.MaterialCardView>

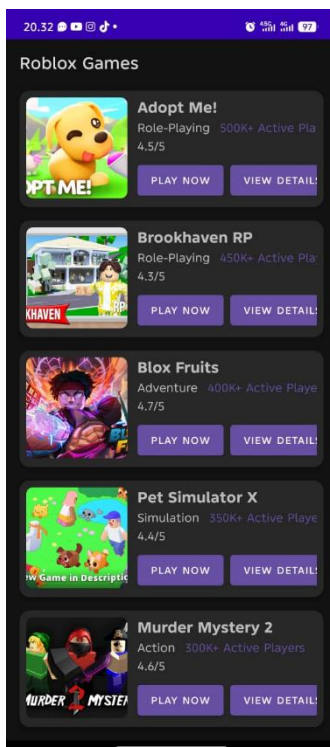
```

Tabel 9 Source Code Jawaban Soal 1

## B. Output Program



Gambar 3 Screenshot Hasil Jawaban Soal 1 Compose



Gambar 4 Screenshot Hasil Jawaban Soal 1 XML

### **C. Pembahasan**

Dalam MainActivity, metode onCreate digunakan untuk menetapkan konten tampilan aplikasi saat aktivitas dibuat. Tema aplikasi ditentukan dengan memanggil AffirmationsTheme, dan tampilan dasarnya dibungkus oleh komponen Surface yang mengisi seluruh ukuran layar serta menerapkan warna latar belakang dari skema warna tema. Di dalamnya, sebuah pengontrol navigasi dibuat menggunakan rememberNavController, kemudian disusun struktur navigasi dengan NavHost. Navigasi dimulai dari layar daftar game, yang ditentukan oleh rute Screen.GameList.route. Saat pengguna menavigasi ke layar detail game, aplikasi akan memuat GameDetailScreen dengan mengambil argumen berupa ID game. Argumen ini diambil dari entri tumpukan balik dan diberi nilai default nol jika tidak ditemukan. Navigasi antara layar-layar tersebut memungkinkan aplikasi menampilkan daftar game dan menampilkan detail dari masing-masing game berdasarkan ID yang dipilih.

NavGraph.kt, Sealed class digunakan untuk mendefinisikan semua rute yang ada di aplikasi. Setiap objek di dalamnya mewakili satu halaman (screen) dengan properti route. object GamesList disini mewakili halaman daftar game, dengan rute "games\_list". Object NavArguments merupakan objek untuk mendefinisikan nama argumen yang digunakan dalam navigasi, agar konsisten dan tidak typo. Terakhir 'GAME\_ID' yang merupakan konstanta string "gameId" yang digunakan sebagai nama parameter pada rute detail game.

Pada RobloxGame.kt, ini merupakan kode yg berfungsi sebagai sumber data dummy di dalam aplikasi. Pertama, ia mendefinisikan sebuah data class bernama RobloxGame yang berfungsi sebagai "cetakan" atau template untuk menyimpan detail informasi sebuah game. Kemudian, ia membuat sebuah object bernama RobloxGamesData yang bertindak sebagai wadah terpusat. Di dalam wadah ini, dibuatlah sebuah daftar statis bernama games yang diisi dengan beberapa objek RobloxGame konkret.

Pad GameDetail, implementasi layar detail game pada aplikasi Android menggunakan Jetpack Compose. Fungsi utama GameDetailScreen menerima ID game dan NavController sebagai parameter. Fungsi ini akan mencari data game yang sesuai dari daftar RobloxGamesData. Jika data ditemukan, layar akan ditampilkan dengan Scaffold yang memiliki TopAppBar berisi judul game dan tombol kembali yang memanfaatkan navigasi ke layar sebelumnya. Bagian utama layar diisi oleh kolom yang dapat di-scroll secara vertikal. Di bagian atas terdapat gambar utama game yang ditampilkan menggunakan AsyncImage dengan efek gradasi gelap di bagian bawahnya agar teks judul dan jumlah pemain terlihat jelas di atas gambar.

### **D. Tautan Git**

Berikut adalah tautan untuk source code yang telah dibuat.

<https://github.com/NaufalElyzar11/PraktikumMobile/tree/main/Modul%203>

## SOAL 2

Mengapa RecyclerView masih digunakan, padahal RecyclerView memiliki kode yang panjang dan bersifat boiler-plate, dibandingkan LazyColumn dengan kode yang lebih singkat?

### **A. Pembahasan**

Pertama, banyak proyek Android yang sudah ada (legacy projects) dibangun menggunakan sistem UI berbasis View System tradisional, di mana RecyclerView menjadi komponen utama untuk menampilkan daftar besar data secara efisien. Migrasi penuh ke Jetpack Compose tidak selalu memungkinkan secara langsung karena bisa berdampak besar pada stabilitas, timeline proyek, dan sumber daya tim.

Kedua, RecyclerView memiliki ekosistem yang sangat matang, dengan dukungan luas untuk berbagai jenis layout seperti GridLayoutManager, StaggeredGridLayoutManager, dan fitur-fitur lanjutan seperti animasi item, drag & drop, swipe, serta integrasi dengan Paging, DiffUtil, dan ViewHolder. Hal-hal ini masih belum sepenuhnya setara dari sisi fleksibilitas dan kontrol jika dibandingkan dengan LazyColumn.

Dengan kata lain, RecyclerView tetap relevan karena stabilitas, fleksibilitas, dan adopsinya yang luas di industri, meskipun Jetpack Compose dengan LazyColumn menjadi arah masa depan untuk pengembangan antarmuka Android yang lebih deklaratif dan efisien.